Week 02: Can Machines Think? The Turing Test and Its Critics

Philosophy of Artificial Intelligence

Philosophy of Artificial Intelligence (11118BLG003). Prof. Dr. Utku Kose. Full Professor, Department of Computer Engineering, Süleyman Demirel University, Isparta, Türkiye. Additional affiliations: University of North Dakota (USA), Universidad Panamericana (Mexico City, Mexico), Vel Tech University (Chennai, India). ORCID 0000-0002-9652-6415. E-mail: utkukose@sdu.edu.tr, utku.kose@und.edu, ukose@up.edu.mx, utkukose@gmail.com. This course is updated in line with current developments in the field. First version: September 2026. Last update: October 2026.

Overview

In 1950, Alan Turing replaced the question of whether machines can think with a game of imitation. This week reconstructs his proposal and his replies to objections. It examines critiques that target the sufficiency of behaviour and interprets recent experiments in which language models were judged to be human [1, 2, 3].

Learning outcomes

By the end of the week, students are expected to describe the imitation game and Turing's prediction, and to evaluate at least three of the objections he considered. They should be able to explain why Blockhead and subcognitive questions challenge behavioural tests. Finally, they should be able to interpret the results of recent three-party Turing tests with appropriate statistical care.

Lecture

Replacing the question

Turing found the question of whether machines can think too vague to answer, so he proposed to replace it with a game [1]. An interrogator communicates in writing with two hidden witnesses, one human and one machine, and must decide which is which. If machines play the game so well that interrogators do no better than chance, the original question loses its point. Turing also made a prediction. By about the end of the twentieth century, an average interrogator would have no more than a 70 percent chance of making the right identification after five minutes of questioning.

The proposal is behavioural and deliberately so. It sets aside the appearance of the machine and the material it is made of, and focuses on verbal behaviour. Turing treated such behaviour as a sufficient basis for attributing thought, just as people attribute thought to one another.

Objections and replies

Turing anticipated nine objections to his view [1]. They are the theological objection, the heads-in-the-sand objection, the mathematical objection, the argument from consciousness and arguments from various disabilities. The list continues with Lady Lovelace's objection, the argument from the continuity of the nervous system, the argument from the informality of behaviour and the argument from extrasensory perception. Three of them return later in the course. The mathematical objection draws on Gödel's theorem and is examined in Week 9. The argument from consciousness, which holds that a machine must feel what it writes, is examined in Weeks 10 and 11. Lady Lovelace's objection, that a machine can only do what it is ordered to do, is examined in Week 13. To the argument from consciousness, Turing replied that it leads to solipsism, since the only way to be sure that a person thinks would be to be that person. Saygin, Cicekli and Akman reviewed the first fifty years of debate about the test and its variants [4].

Is behaviour enough?

Critics have questioned whether passing the test is sufficient for thought. Block imagined a machine, later called Blockhead, that stores a sensible reply for every possible conversation up to the length of the test [2]. Such a lookup table would pass, yet it would be as unintelligent as a jukebox. Block concluded that intelligence depends on how behaviour is produced, not only on the behaviour itself, a position he called psychologism. The practical impossibility of the table does not answer the point, since the argument concerns what the test measures, not what can be built.

French argued that the test is too demanding in a different way [5]. Subcognitive questions probe associations that only an entity with a human life history would share. An example is asking how good a made-up word would be as a name for a breakfast cereal. The test may therefore detect humanness rather than intelligence. Weizenbaum's ELIZA showed a third problem from the opposite direction: A simple program that transformed user sentences with pattern-matching rules led some users to attribute understanding to it [6]. Human judges are not neutral instruments.

The test in the age of language models

Jones and Bergen ran randomised, controlled and pre-registered three-party Turing tests in which interrogators held five-minute conversations simultaneously with a human and a machine [3]. When prompted to adopt a humanlike persona, GPT-4.5 was judged to be the human in 73 percent of the games, significantly more often than the real human participants. LLaMa-3.1-405B with the same prompt was judged human 56 percent of the time, while ELIZA and GPT-4o without a persona reached 23 and 21 percent. Figure 2.1 shows these reported rates.

The results meet Turing's operational criterion in its standard form. Whether they show that the systems think is exactly the question that Block and French raised. The results also depended strongly on the persona prompt, which suggests that the test measures skill at a particular social

performance. The interactive lab asks students to act as interrogators, to compute whether their own accuracy exceeds chance and to estimate the size of Blockhead's table.

> ***Pause and reflect.*** *If a system passes the test only with a carefully written persona, who or what has passed the test: the model, the prompt writer or the combination?*

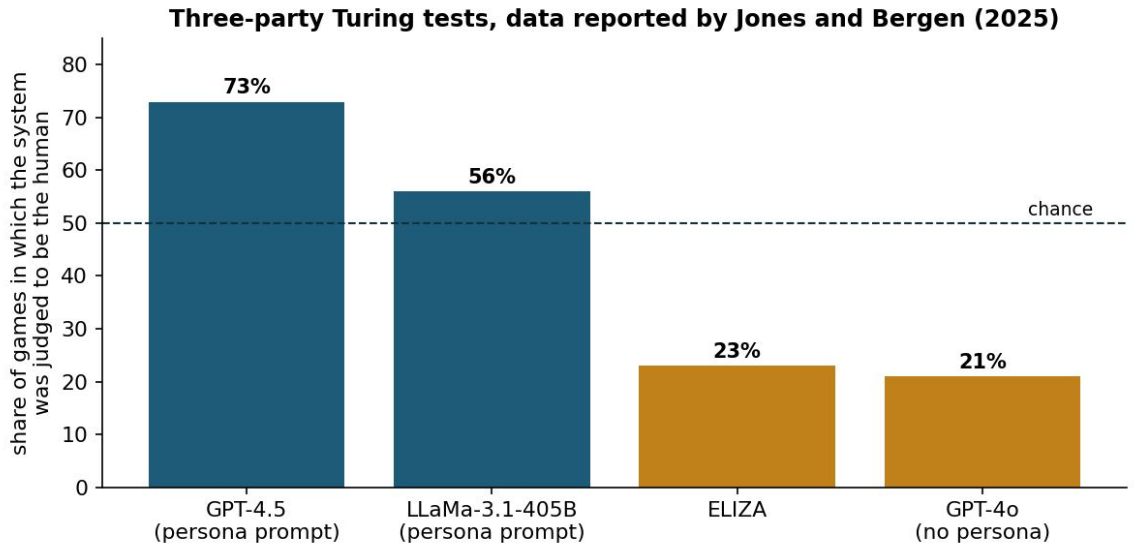


Figure 2.1. Share of games in which each system was judged to be the human in the three-party Turing tests reported by Jones and Bergen; the dashed line marks chance [3].

Hands-on components

Interactive lab: Part A presents eight short replies, constructed for this exercise, each written either by a person or in the style of a language model. Guess the source of each reply and test whether the accuracy exceeds chance. Part B estimates the size of Blockhead's lookup table. Part C tests the consistency of a position on behavioural tests [1, 2]. Link: https://utkukose.github.io/philosophy-of-ai-NB-lecture/weeks/week-02/lab.html

Colab notebook: It implements a small ELIZA-style program and models interrogators with signal detection theory, which relates their accuracy to Turing's 70 percent criterion. It then tests judge accuracy against chance and computes the size of Blockhead's table [1, 2, 6]. Link: https://colab.research.google.com/github/utkukose/philosophy-of-ai-NB-lecture/blob/main/weeks/week-02/NB02_turing_test.ipynb

Reflection prompts

1. How accurate were you in Part A, and what cues did you rely on? Would the same cues work against a persona-prompted model?
2. Does Blockhead refute behavioural tests of intelligence, or only show their limits? Defend your view.
3. Design a question that you believe a present language model would answer differently from a person. Explain why.

Weekly task and submission

What to do. Write an argument analysis of Turing's proposal. First, reconstruct Turing's argument as numbered premises and a conclusion. Second, state which premise Block attacks with Blockhead and which premise French attacks with subcognitive questions. Third, assess whether the three-party tests of Jones and Bergen change the debate.

Where to look. Lecture: sections "Replacing the question", "Is behaviour enough?" and "The test in the age of language models". Interactive lab: Part B for the size of Blockhead's table and Part C for testing your own position. Colab notebook: Section 2 for Turing's 70 percent criterion, Section 3 for testing judges against chance and Section 4 for the size of Blockhead.

Example. One premise could read: People attribute thought to one another on the basis of verbal behaviour alone. The argument from consciousness attacks the step from this premise to machines, and Turing replied that it leads to solipsism. Block and French attack other premises, which your analysis must identify.

What to submit. About 500 words, and the Colab notebook with Exercises 1 and 2 completed.

The weekly task supports self-learning and builds a personal portfolio. In an active semester, students who follow the course with the instructor can send their work for evaluation to utkukose@sdu.edu.tr or utkukose@gmail.com. The e-mail contains the items that the task lists under What to submit. It also includes the learning log exported from the interactive lab (tab: Reflect and submit).

References

[1] Turing, A. M. (1950). Computing machinery and intelligence. Mind, 59(236), 433-460. https://doi.org/10.1093/mind/LIX.236.433

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[3] Jones, C. R., & Bergen, B. K. (2025). Large language models pass the Turing test. arXiv preprint arXiv:2503.23674. https://arxiv.org/abs/2503.23674

[4] Saygin, A. P., Cicekli, I., & Akman, V. (2000). Turing test: 50 years later. Minds and Machines, 10(4), 463-518.

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